

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Outer Barrier Valve & Speed Controller Upgrade (S/N:8007 & Below)	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	2 Oct 2016	First Release	ONG KEAN SHENG

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Parts requirement :

Part Number	Part Description	Qty
2860-0003	TUBING, TU POLYURETHANE, 6MM	1 Roll
2820-0117	FITTING,FLOW CONTROL,D6.0	2
4XASP-A154	S2EX Outer barrier Solenoid Valve kit	1

Parts removal

1. Close SSCA
2. Close X ray Source at Operator Console Panel
3. Unplug Parker Clip Fitting as below Figure 1.

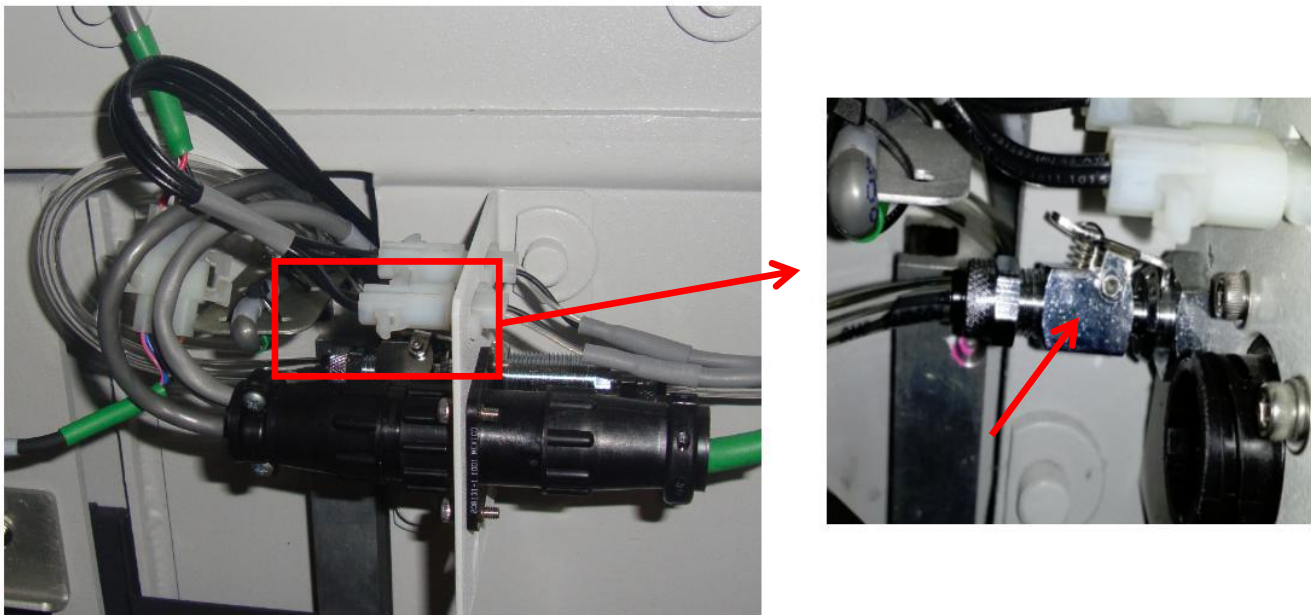


Figure 1

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4. Remove 2 x tubing from outer barrier speed controller. May refer to below Figure 2.



Figure 2

5. Remove 2 x Outer Barrier Speed Controller as below Figure 3



Figure 3

6. Remove 2 x Tubing from XXL Outer Barrier Solenoid Valve (Outlet Port) as below Figure 4.

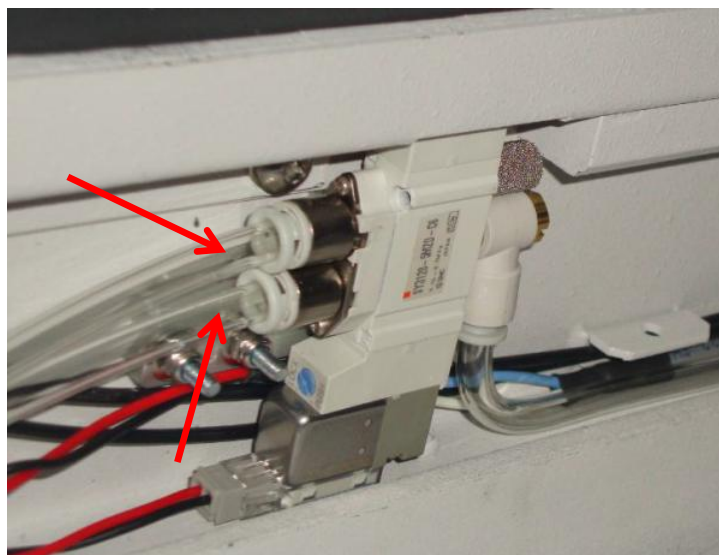


Figure 4

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7. Remove 1 x Tubing from XXL Outer Barrier Solenoid (Inlet Port) as below Figure 5. Remove Input Cable as well.

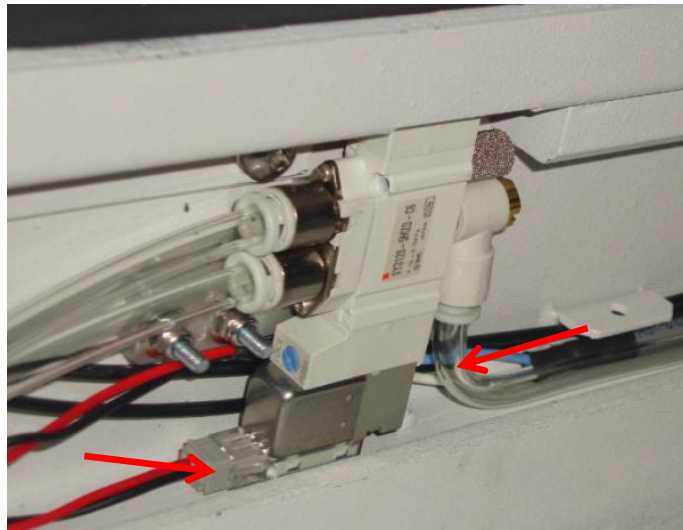


Figure 5

8. Loosen 2 x Nut and remove entire XXL Outer Barrier Solenoid Valve as below Figure 6.

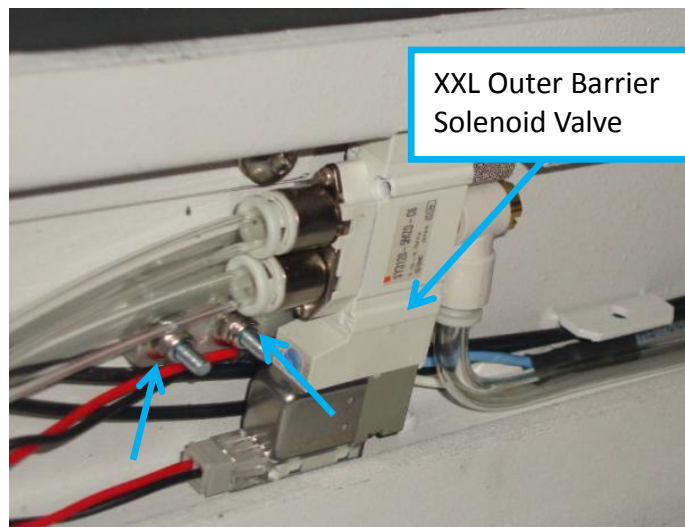


Figure 6

9. Flip the XXL Outer Barrier Solenoid Valve. Loosen 2 x Philip Screws and remove the Holding Bracket. May refer to below Figure 7.



Figure 7

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10. Position Holding Bracket & 2 x Philip screws at visible location. Parts will require to re-use.

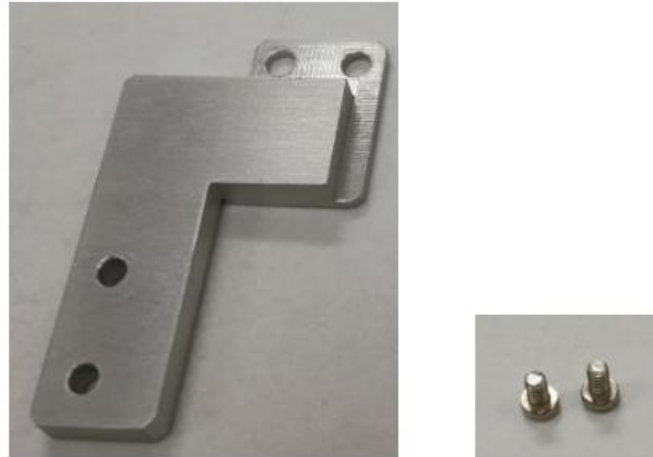


Figure 8

11. Remove entire tubing together with Parker clip fitting. May refer to below Figure 9.

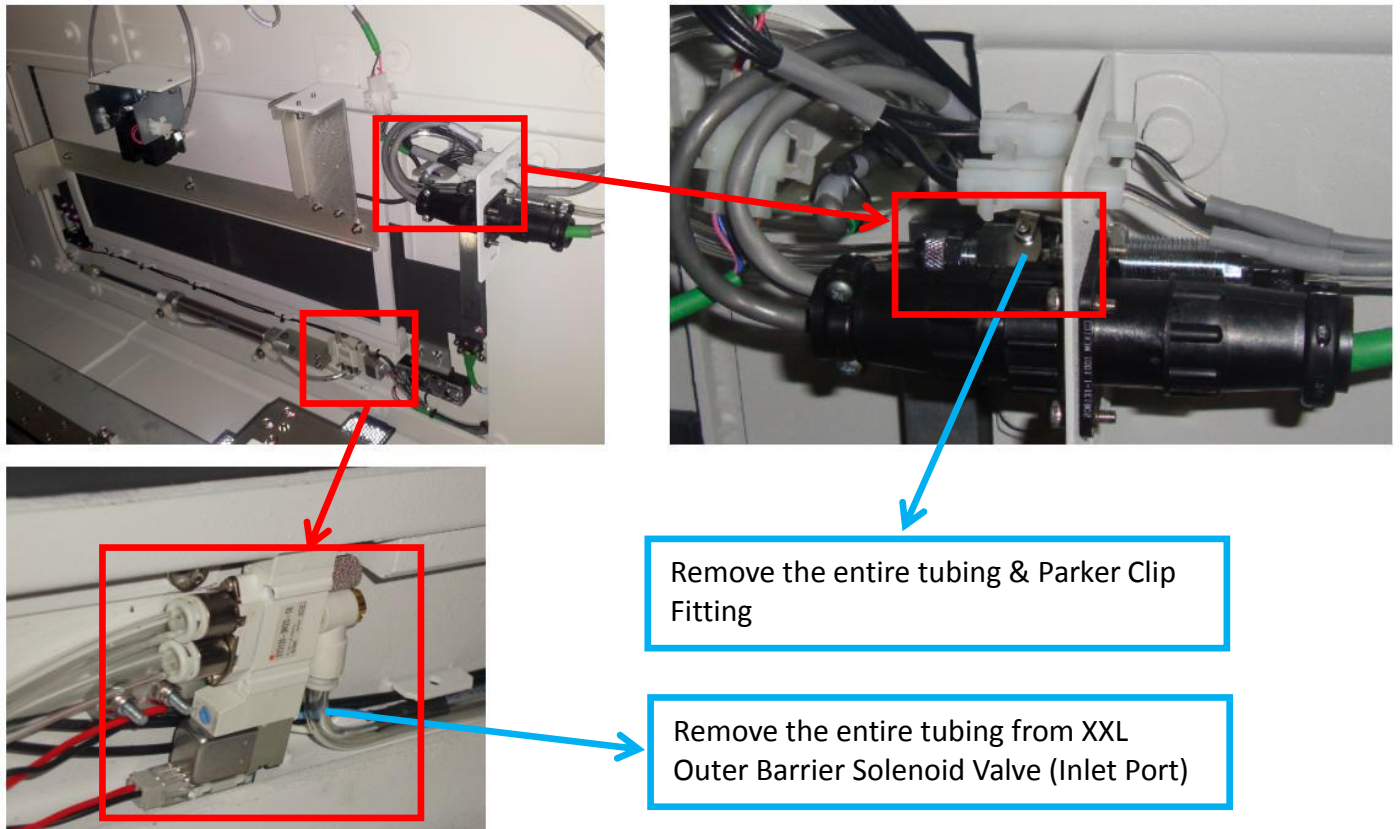


Figure 9

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12. Gently loosen the nut and slide through existing tubing. May refer to below Figure 10.

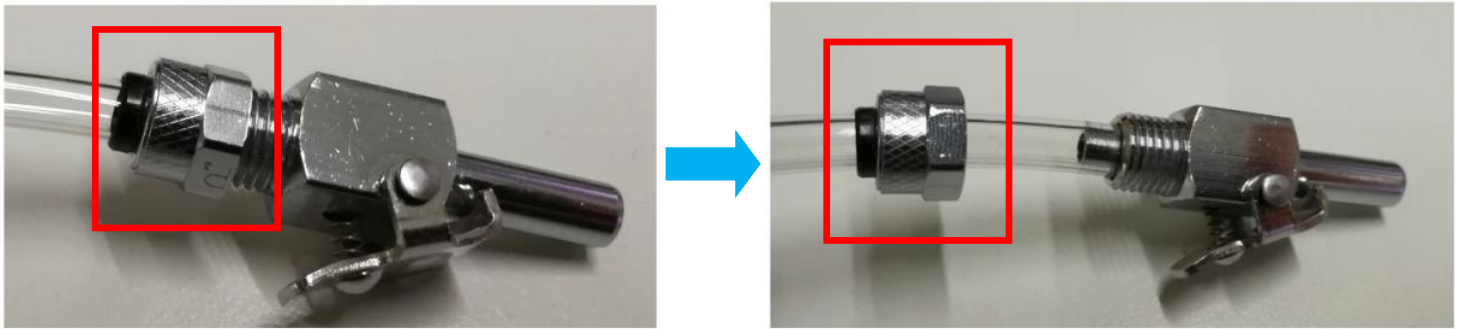


Figure 10

13. Gently remove the existing tubing from Parker Clip Fitting as Figure 11.

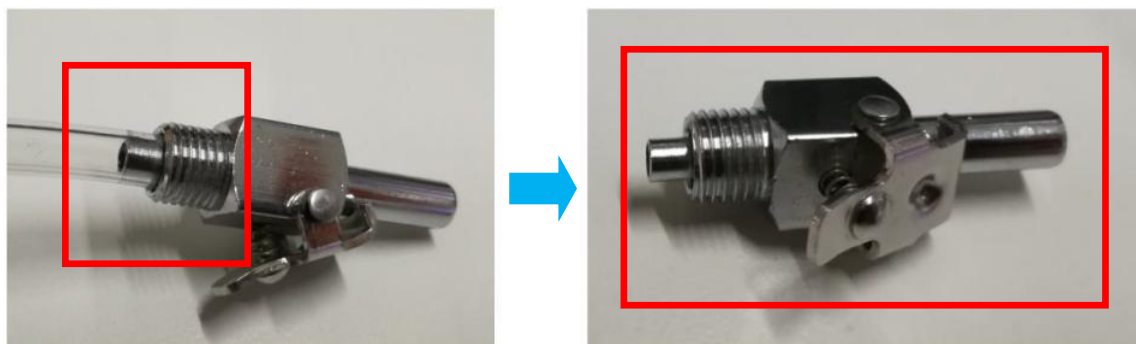


Figure 11

Part replacement

14. Assemble Holding bracket by tighten 2 x Philip Screws at New S2EX Outer Barrier Solenoid Valve Kit (PN : 4XASP-A154)

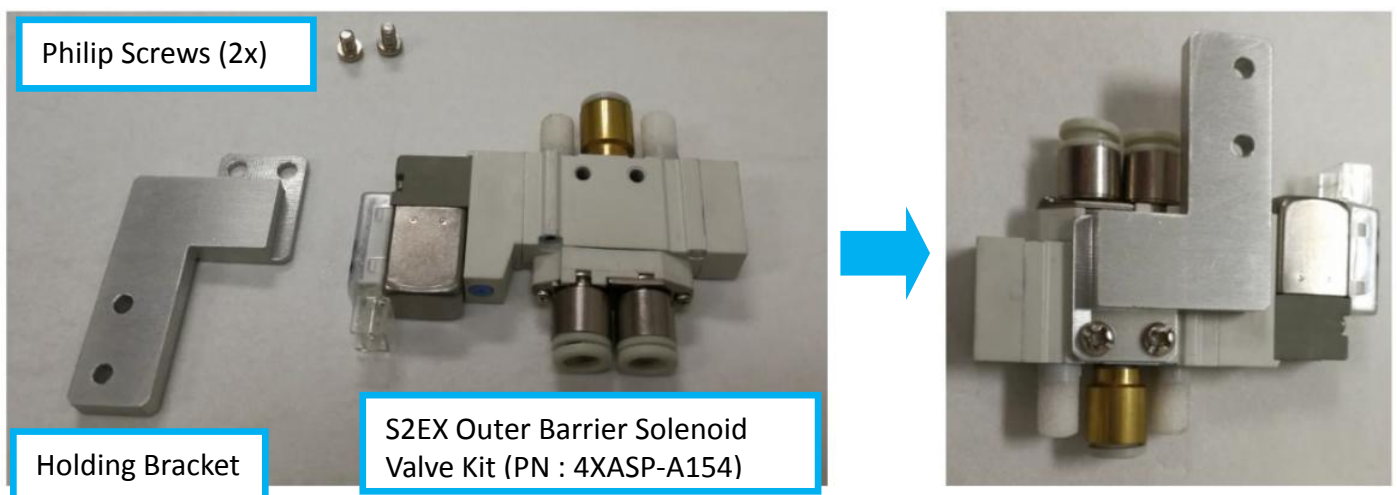


Figure 12

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15. Assemble entire Outer Barrier Valve assembly at Outer Barrier. Tighten 2 x Nut. May refer to Figure 13.

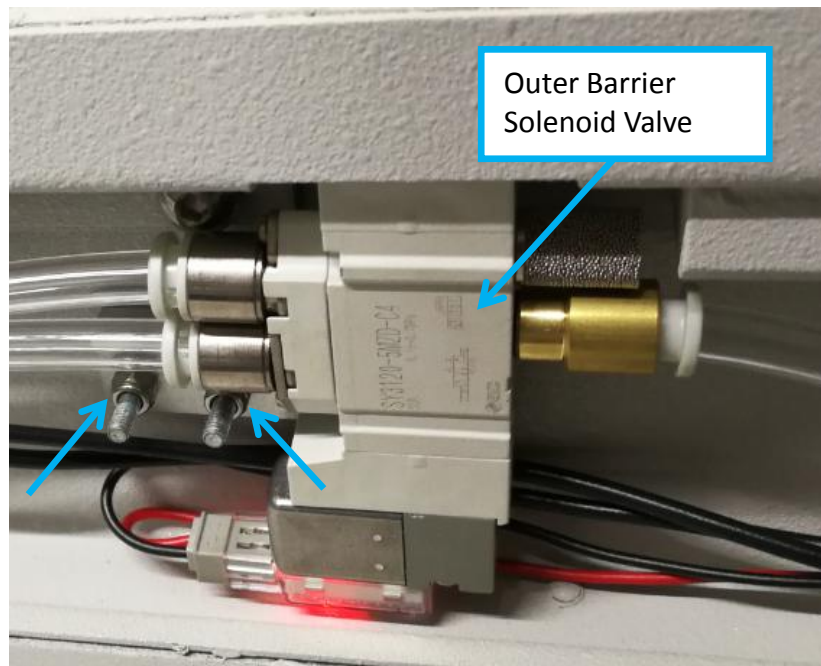


Figure 13

16. Assemble 2 x New Speed Controller (2820-0117) at outer barrier cylinder as Figure 14.

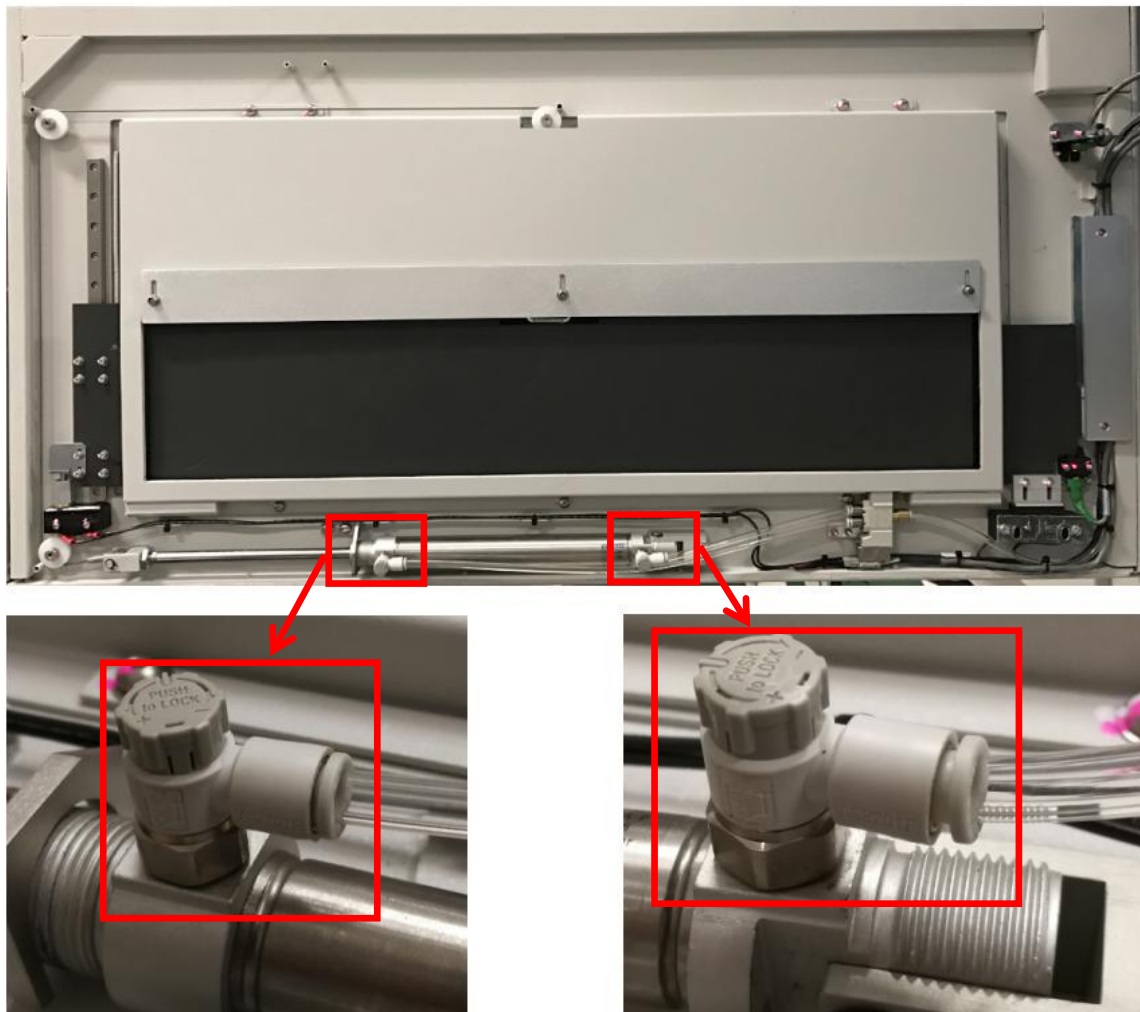


Figure 14

17. Cut 2 x tubing (PN : 2860-0003) into desire length by refer to below Figure 15. Plug 2 x Tubing by refer to below orientation.

Orientation	Routing Path	Cutting Length (Approximately)	Figure
A1-A2	Speed Controller to Outer Barrier Valve	39.5cm	15
B1-B2	Speed Controller to Outer Barrier Valve	18.5cm	16

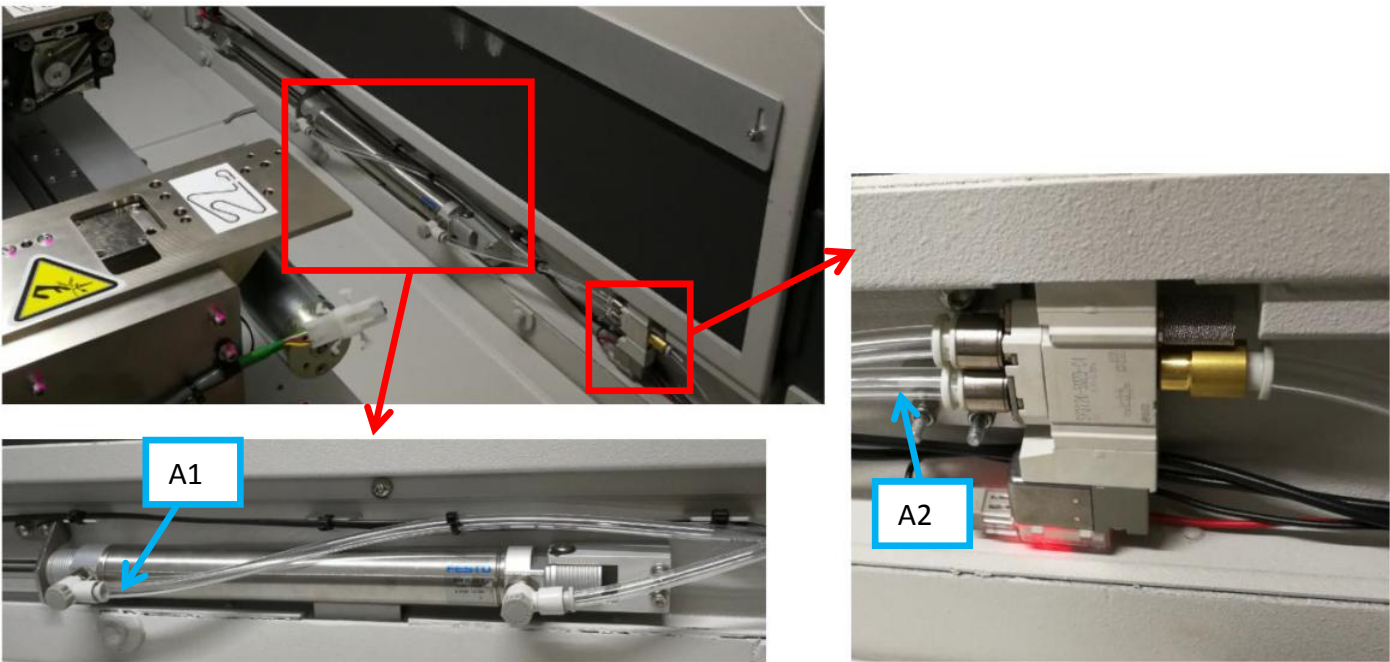


Figure 15

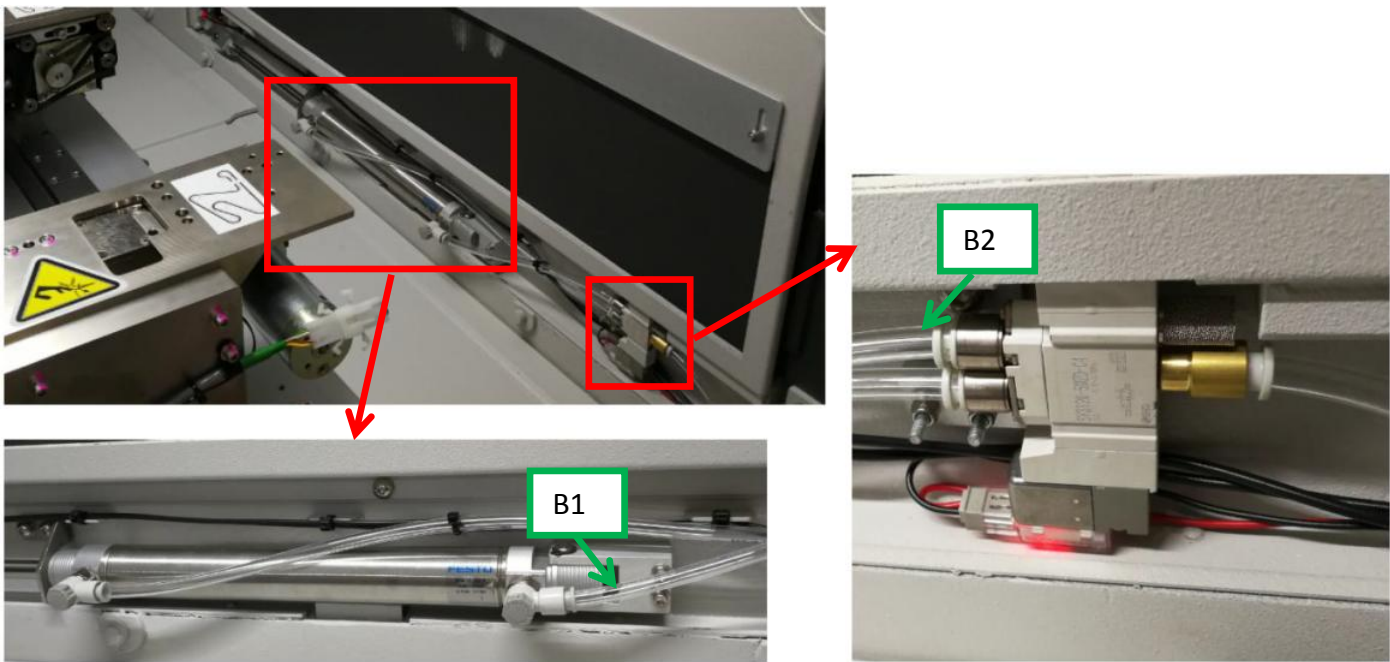


Figure 16

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18. Cut 1 x Tubing (PN : 2860-0003) into desire length (Approximately 85cm) by refer to routing path (Blue Highlight)

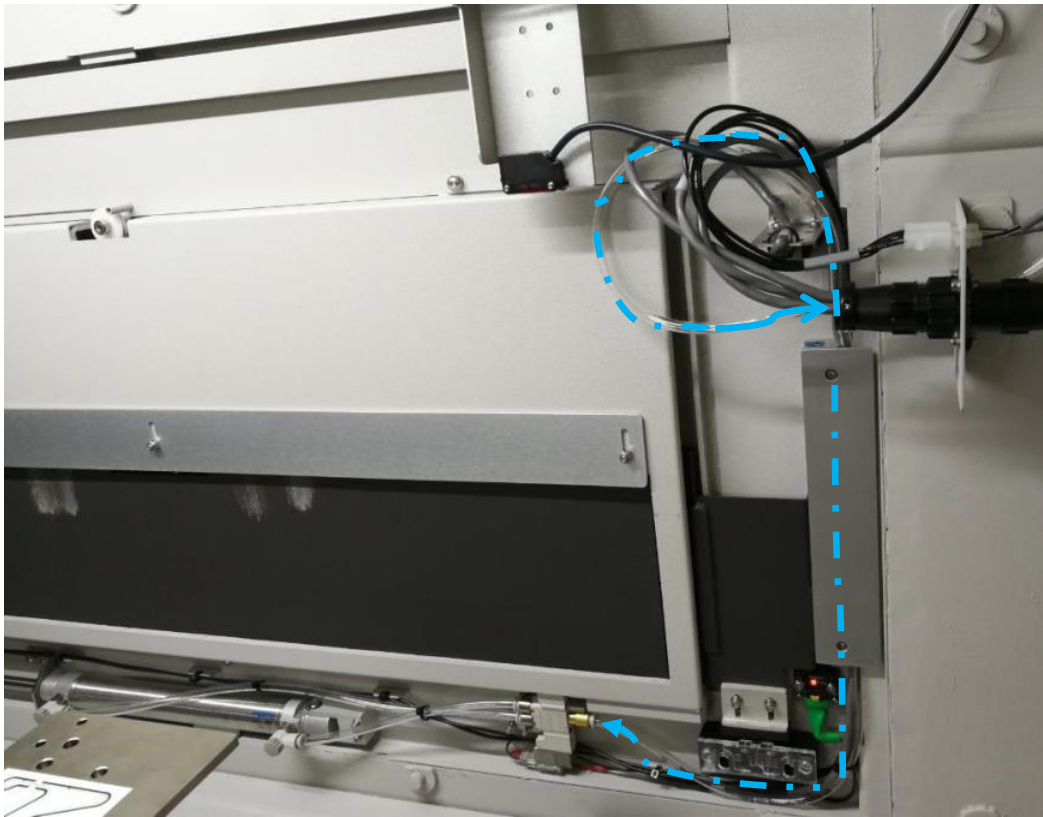


Figure 17

19. Plug the other end of tubing (Approximately 85cm) at Parker Clip Fitting. May refer to below Figure 18.

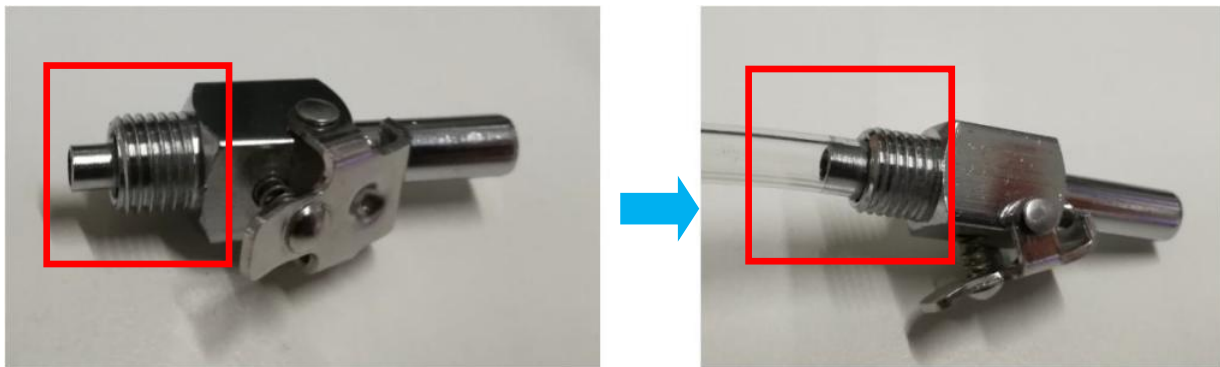


Figure 18

20. Gently slide the nut through tubing. Tighten the nut. May refer to Figure 19.

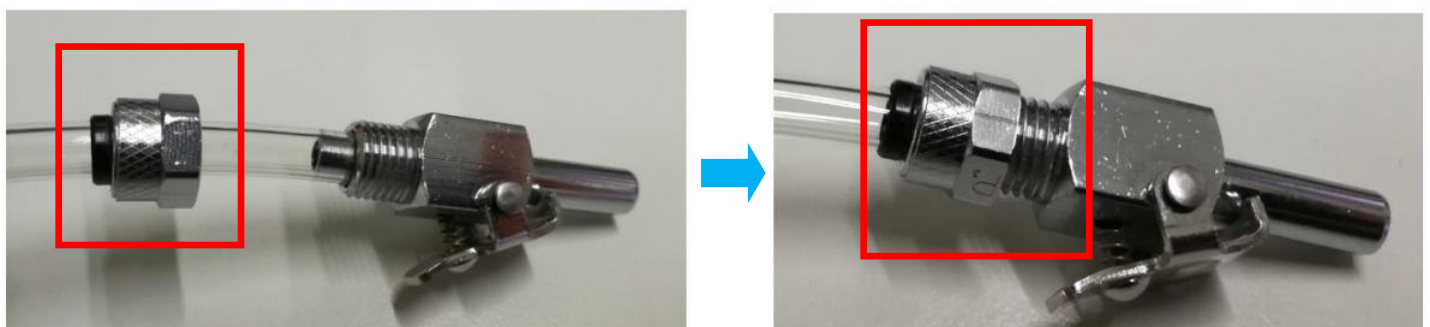


Figure 19

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21. Gently plug the other end of tubing into New Outer Barrier Solenoid Valve (Inlet port) & apply cable tie to secure as Figure 20. Connect Input Cable

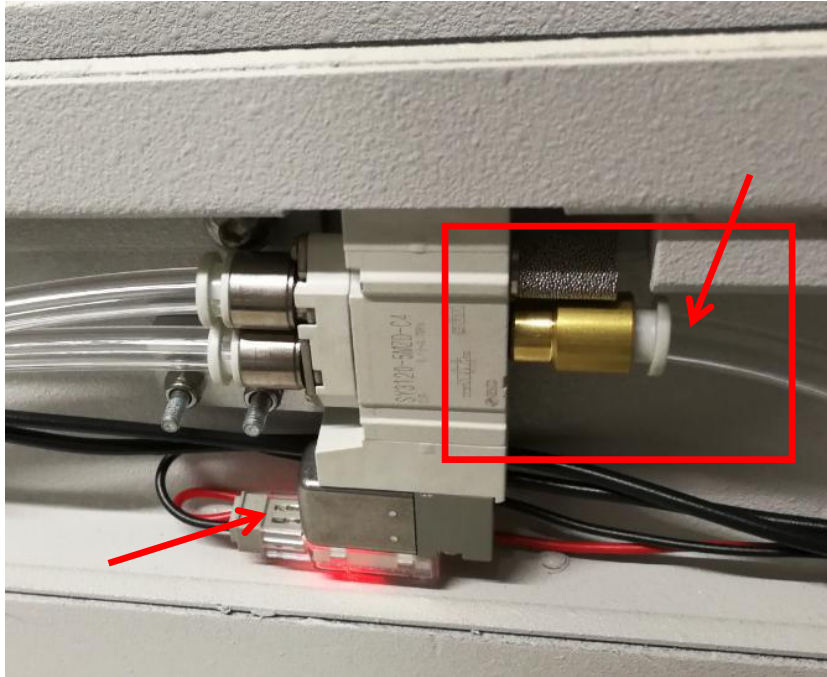


Figure 20

22. Gently route the entire tubing by refer to below route (Blue Highlight). Press & lock the Parker Clip Fitting. May refer to below Figure 21 & 22. Apply necessary cable ties to secure the tubing.

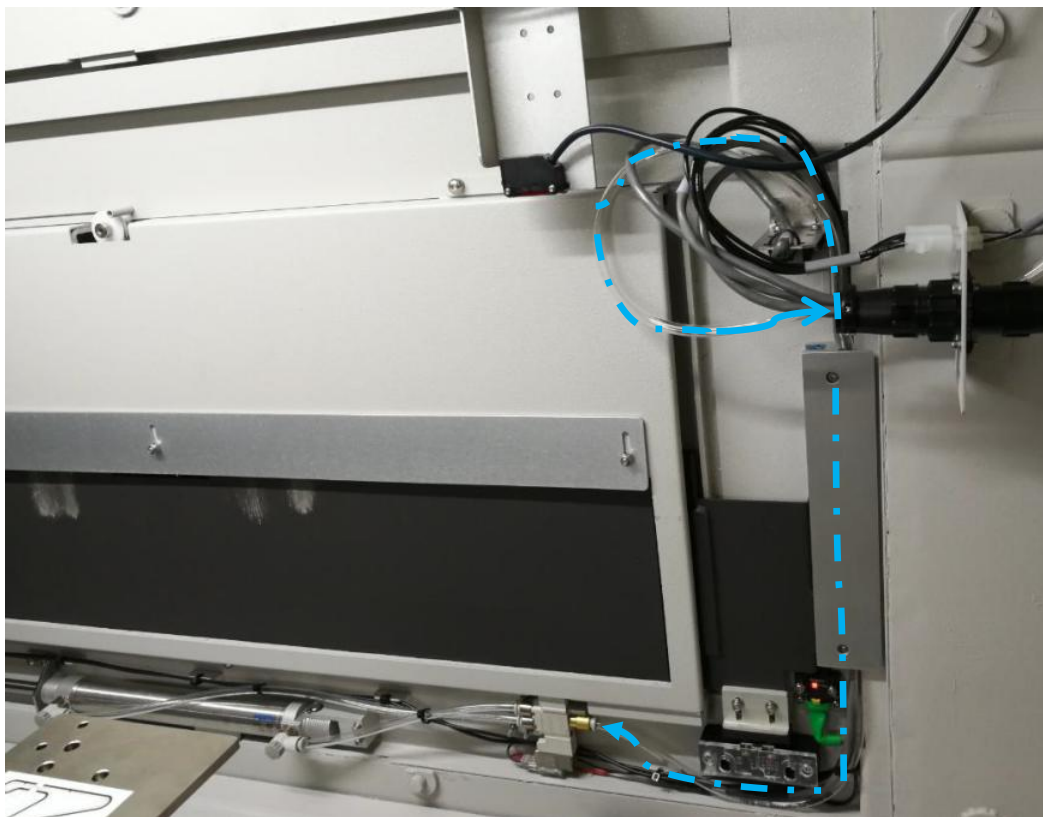


Figure 21

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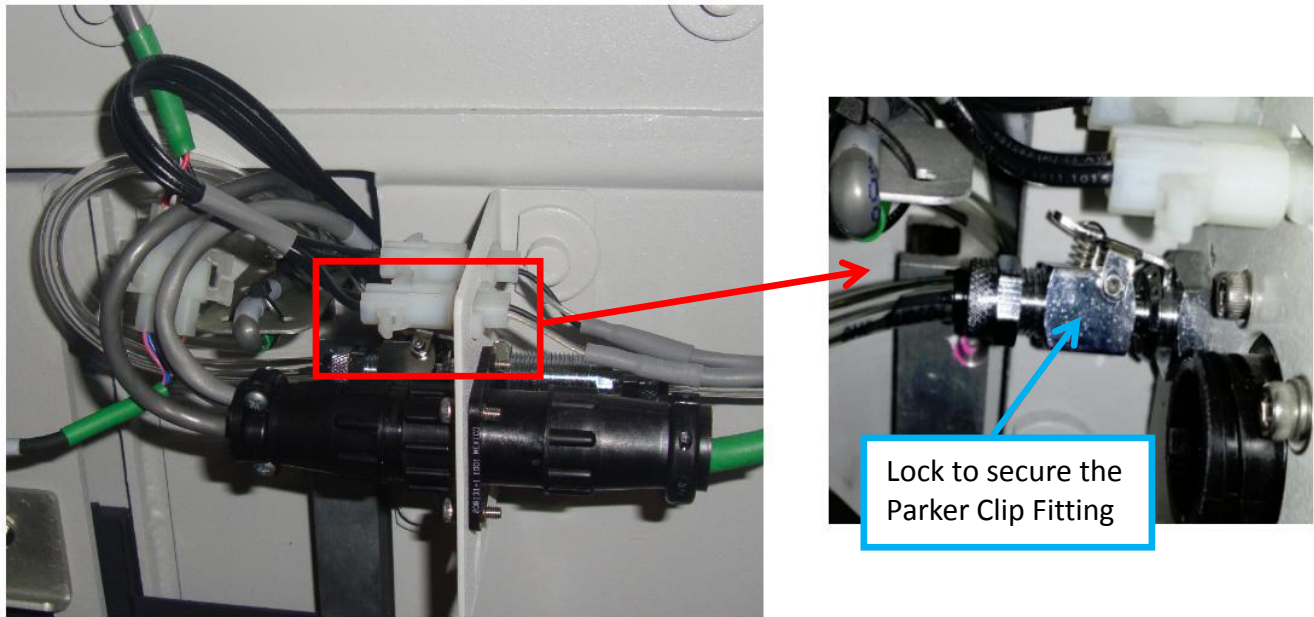


Figure 22

23. Manually move the X Axis Stage towards particular barrier.



Figure 23

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24. Visual check outer barrier tubing whether contain any friction with X Axis Stage.

a) If friction persist, re-arrange tubing position or re-cut new tubing into desire length.

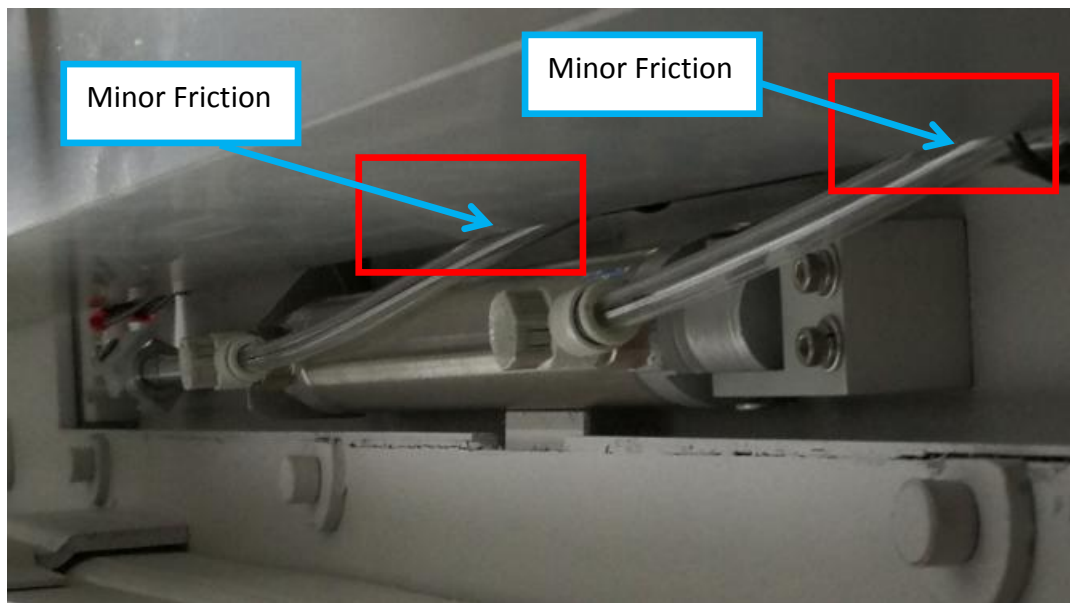


Figure 24

25. Remove all tools from XY Stage

26. Close all access door

27. Open SSCA

28. Open V810 GUI

29. Initialize at Panel handling

30. Open & Close particular Outer barrier which perform Valve & Speed controller upgrading. Record down Opening & Closing duration as below Figure 25.

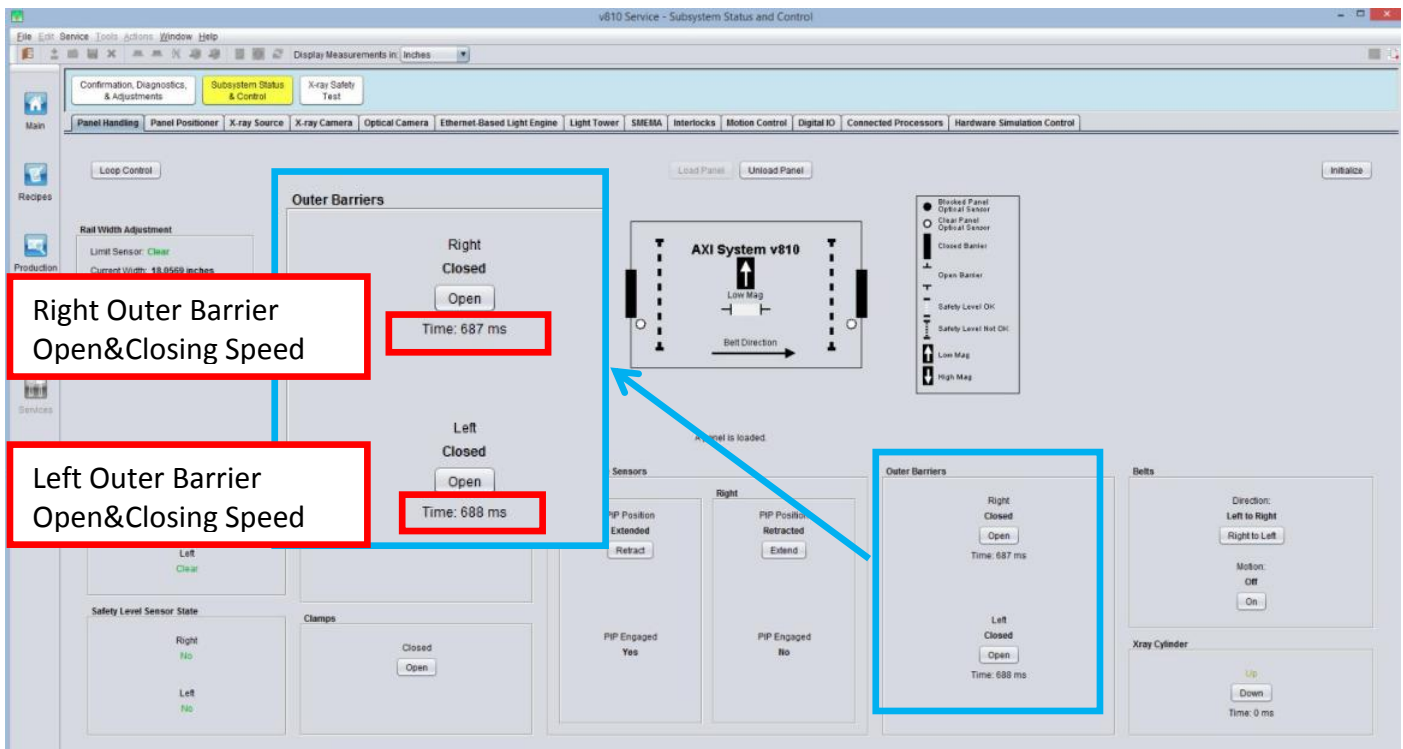


Figure 25 Outer Barrier Opening & Closing Duration

31. Adjust 2 X Speed Controller at Outer Barrier cylinder by refer below table.

Outer Barrier Status	Outer Barrier Speed
Open	1200ms~1500ms
Close	1200ms~1500ms

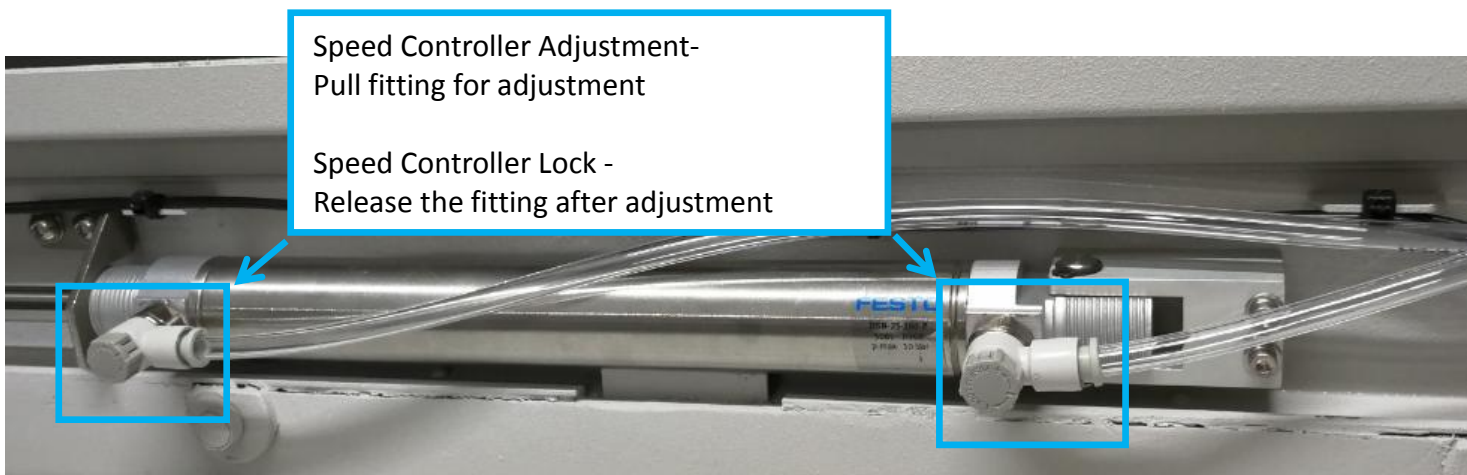


Figure 26